

Financial Bot with LLMs Project



Instructor: Michael Bazzoli

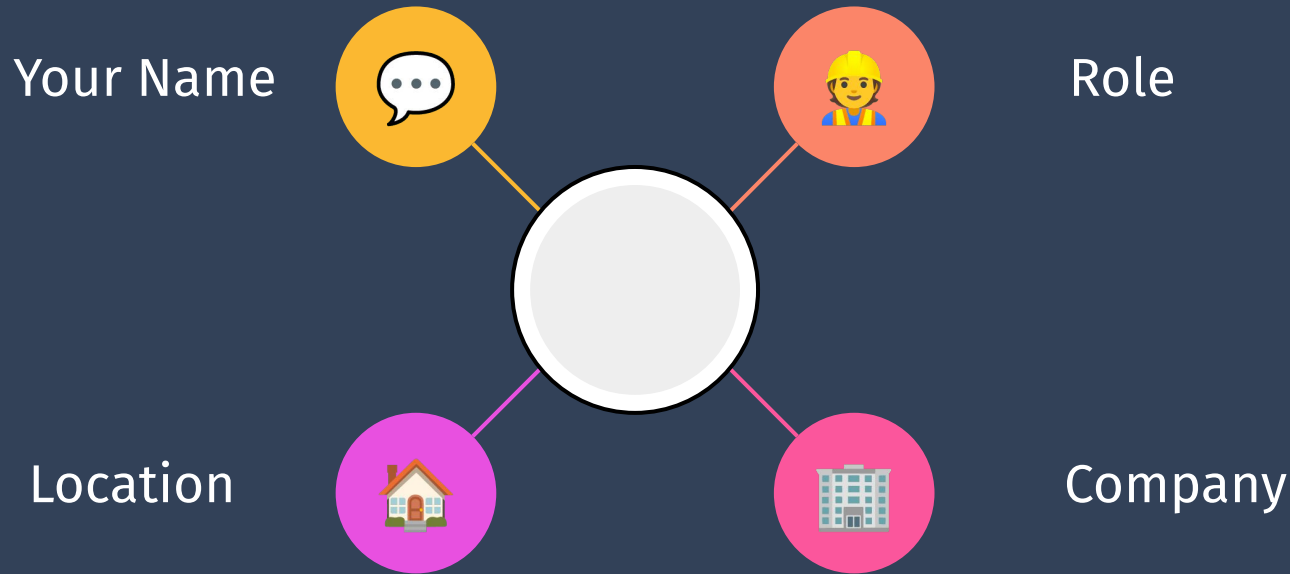
A little about me, Michael Bazzoli

Staff MLE - DataRobot adding GenAI/LLM features
Prev: MLE @ Amazon, Wyze, Microsoft

Computer Engineering –
University of Illinois @ Urbana-Champaign

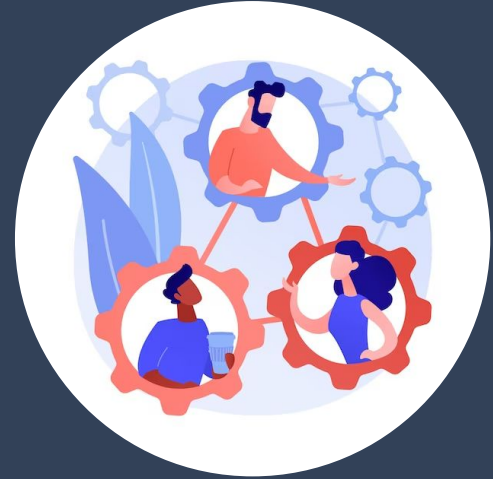
Based in Los Angeles, CA

Welcome to today's class, before we begin, pop into the chat:



Optimize Your Experience

- ✓ Interact with your instructors via Sunday live class.
- ✓ Don't be shy to speak up and get clarifications !



- ✓ Solve your assignments, MCQs and other assessments and get feedback (Thursday review session)
- ✓ Use your resources! It's your experience – what you put in is what you'll get out.

Learning Objectives

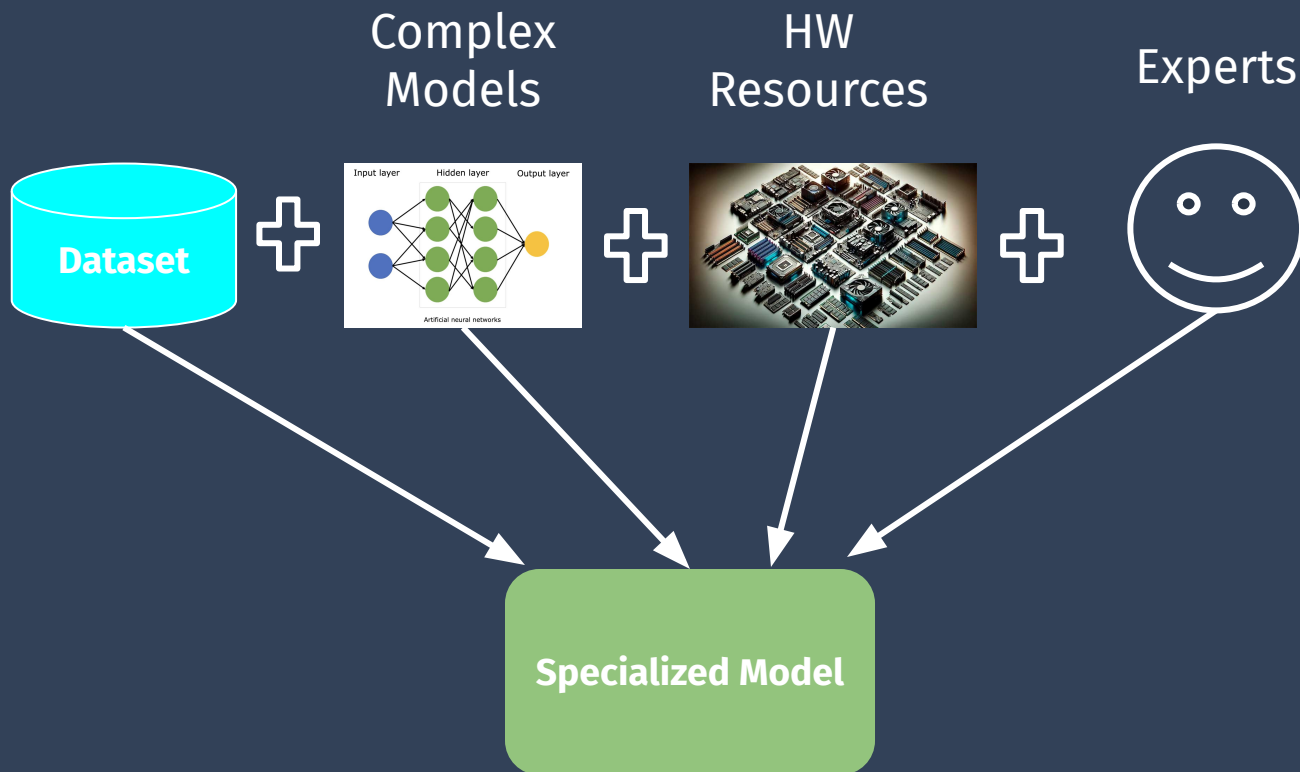
- Gets hands-on experience on building a RAG application.
- Quickly give a reminder on RAG first.

How is Today's Content Relevant to my Role?

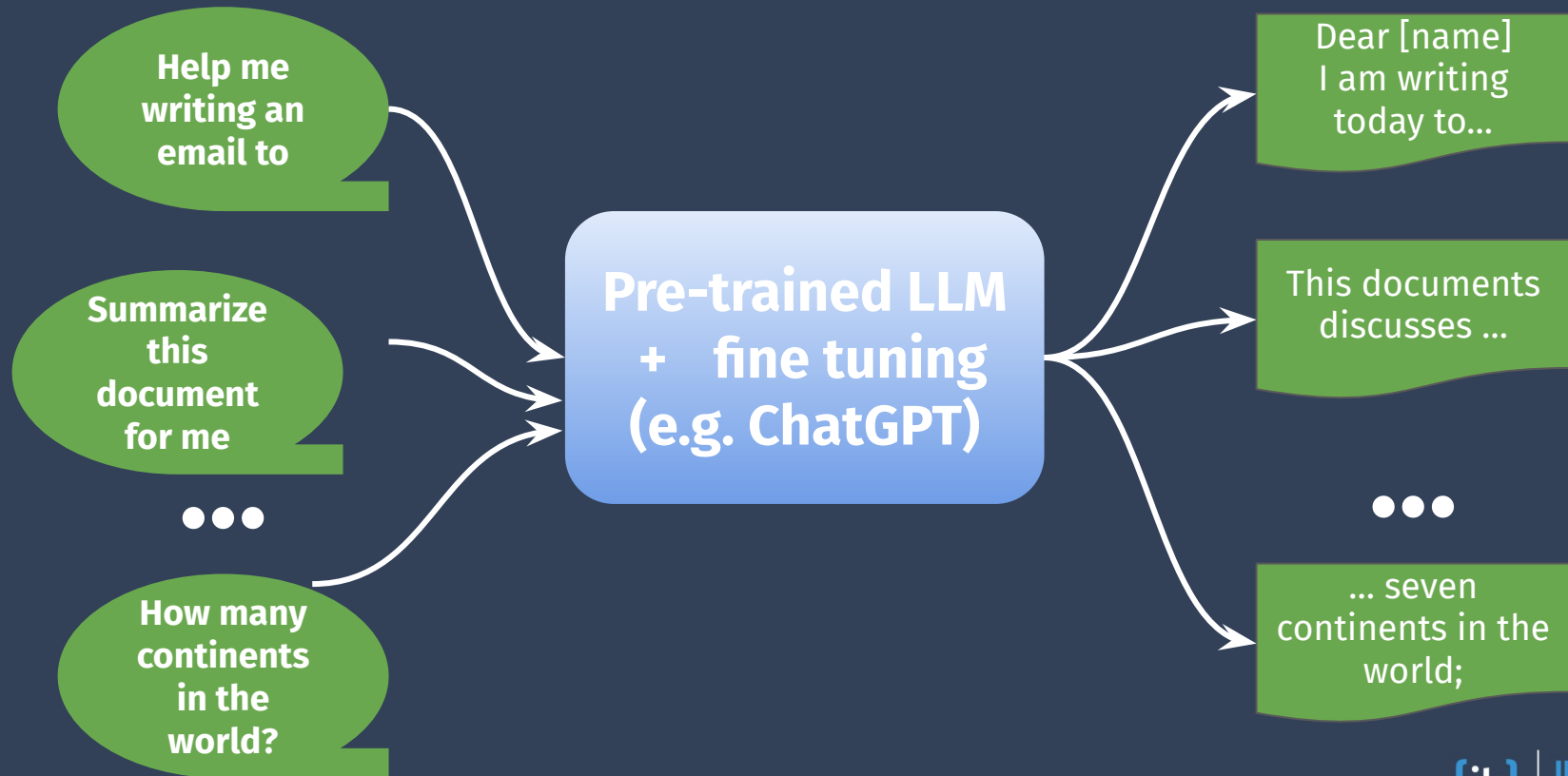
- **PMs:** Understand what types of Generative AI product features could be built with currently available tools.
- **TPMs:** Understand the technical feasibility and requirements of implementing certain generative AI features.
- **SDEs:** Become familiar with how to implement Generative AI capabilities in applications.
- **Engineering Managers:** Become familiar with options available for integrating Generative AI capabilities into the applications that you own.
- **DevOps Engineers:** Get a first look at software and infrastructure requirements for running or accessing Generative AI models.

Training a Specialized Model

What Does it Take?



Large Language Model (LLM)



LLM

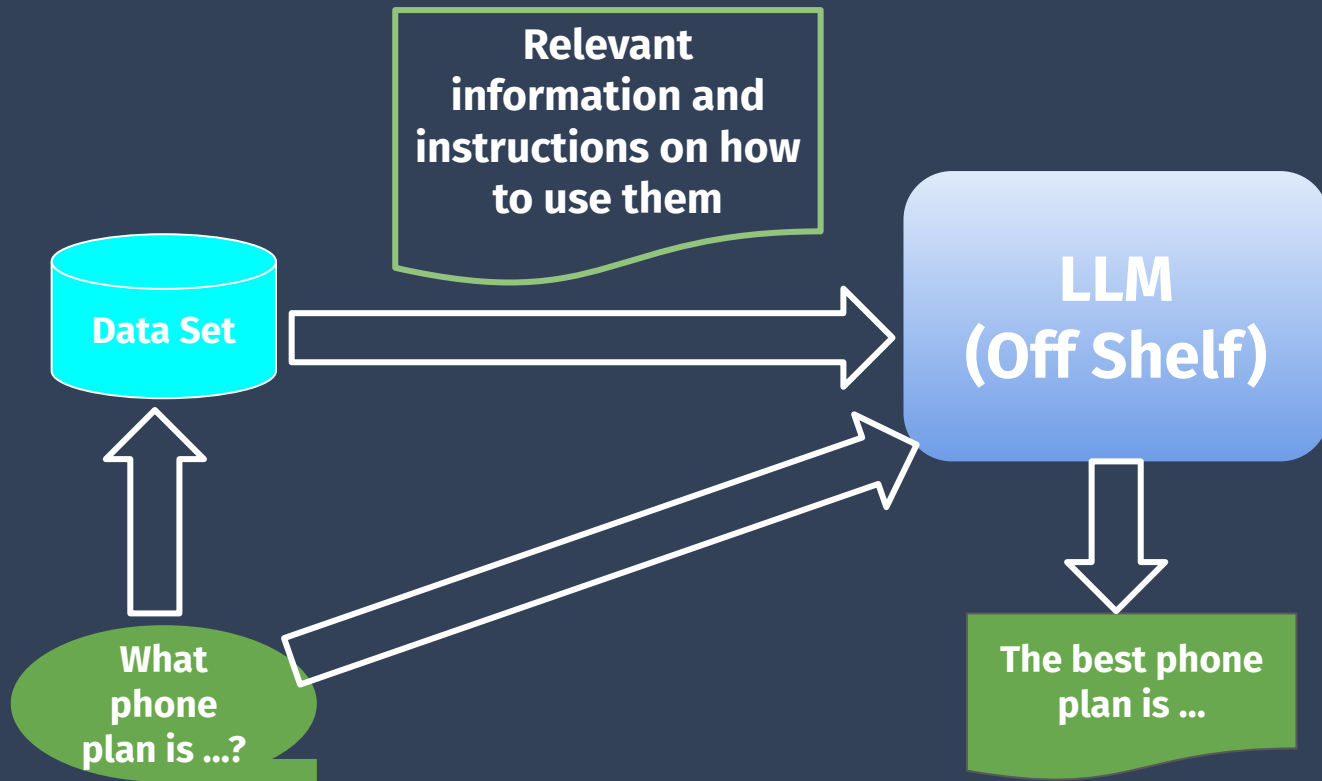
Solved Many Problems

- Pre-trained on huge amount of data
- Fine-tuned on task-specific dataset (instruction following)
- Fine-tuned with RLHF to align with human preferences
- Can be used as is on a variety of use cases.
- Many Cloud APIs are available

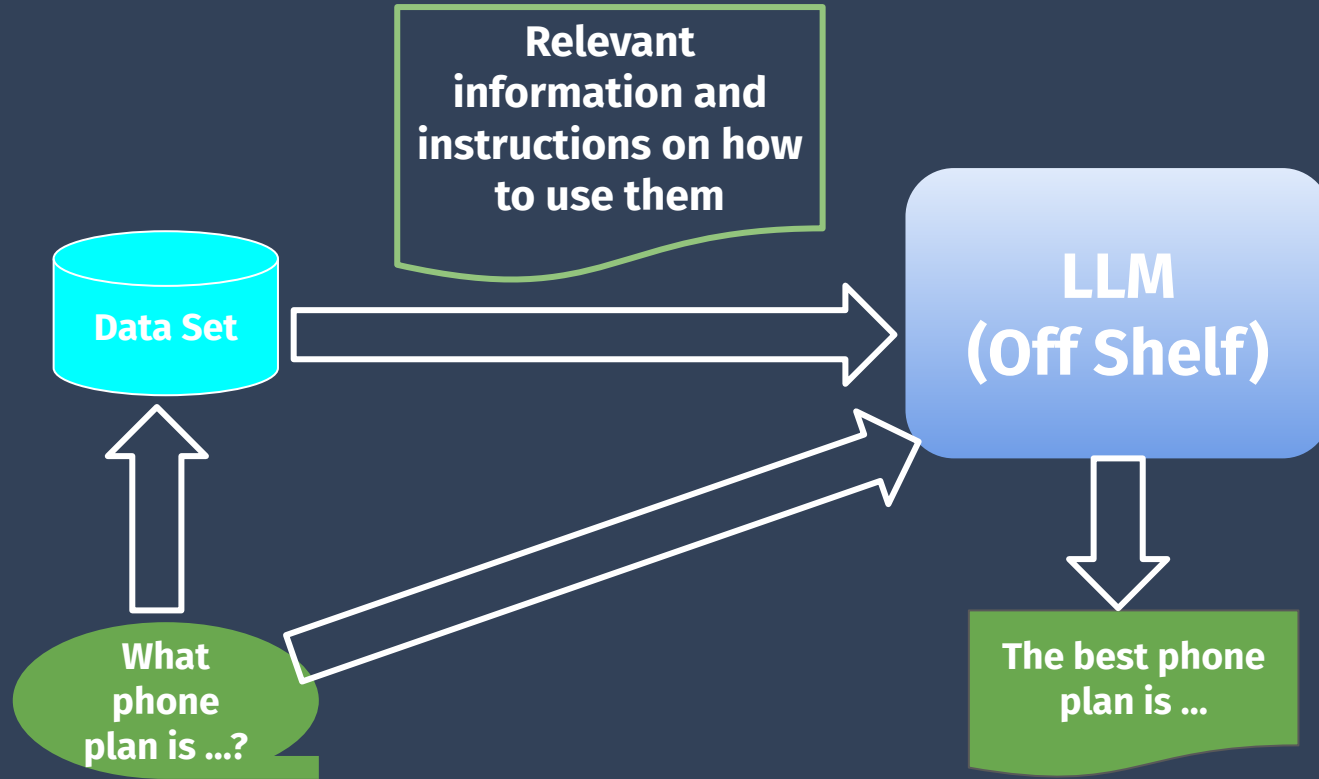
Limitations

- How to use external knowledge (think not publicly available ever or @ time of training)
 - Candidate resumes
 - Internal slack messages and internal technical docs
 - ...
- Knowledge cutoff
 - Who won the last Turing Award?

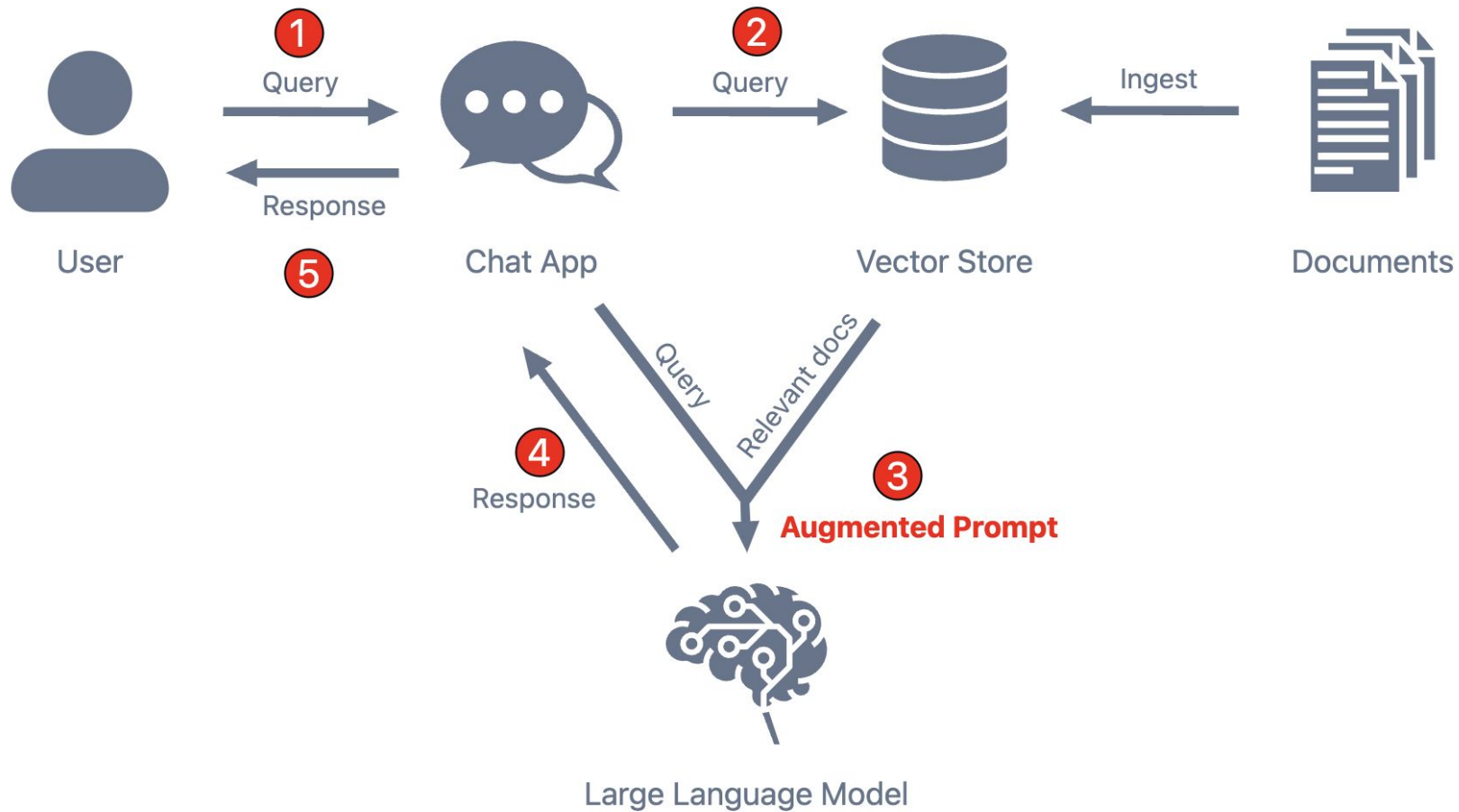
Retrieval Augmented Generation (RAG)

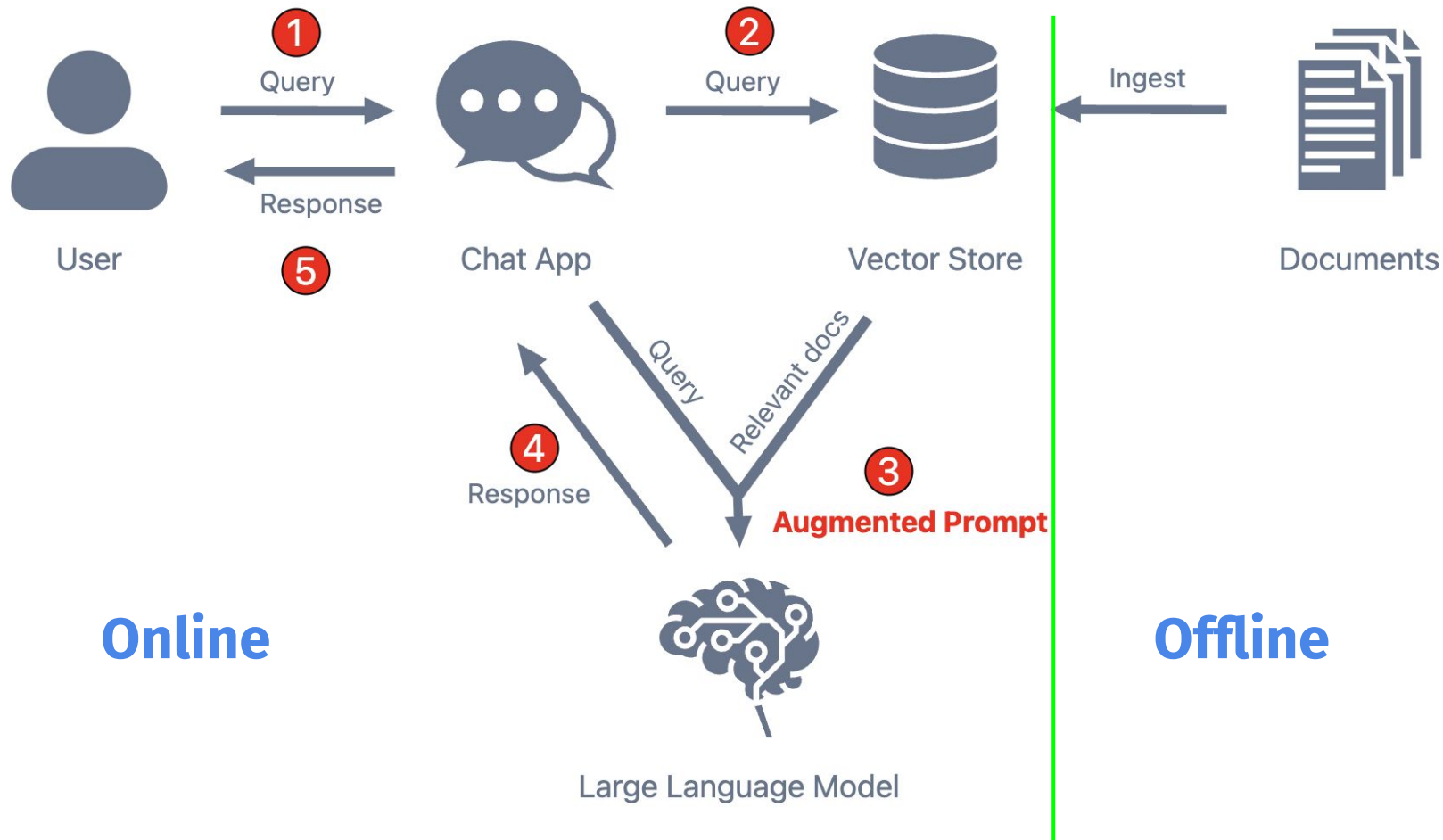


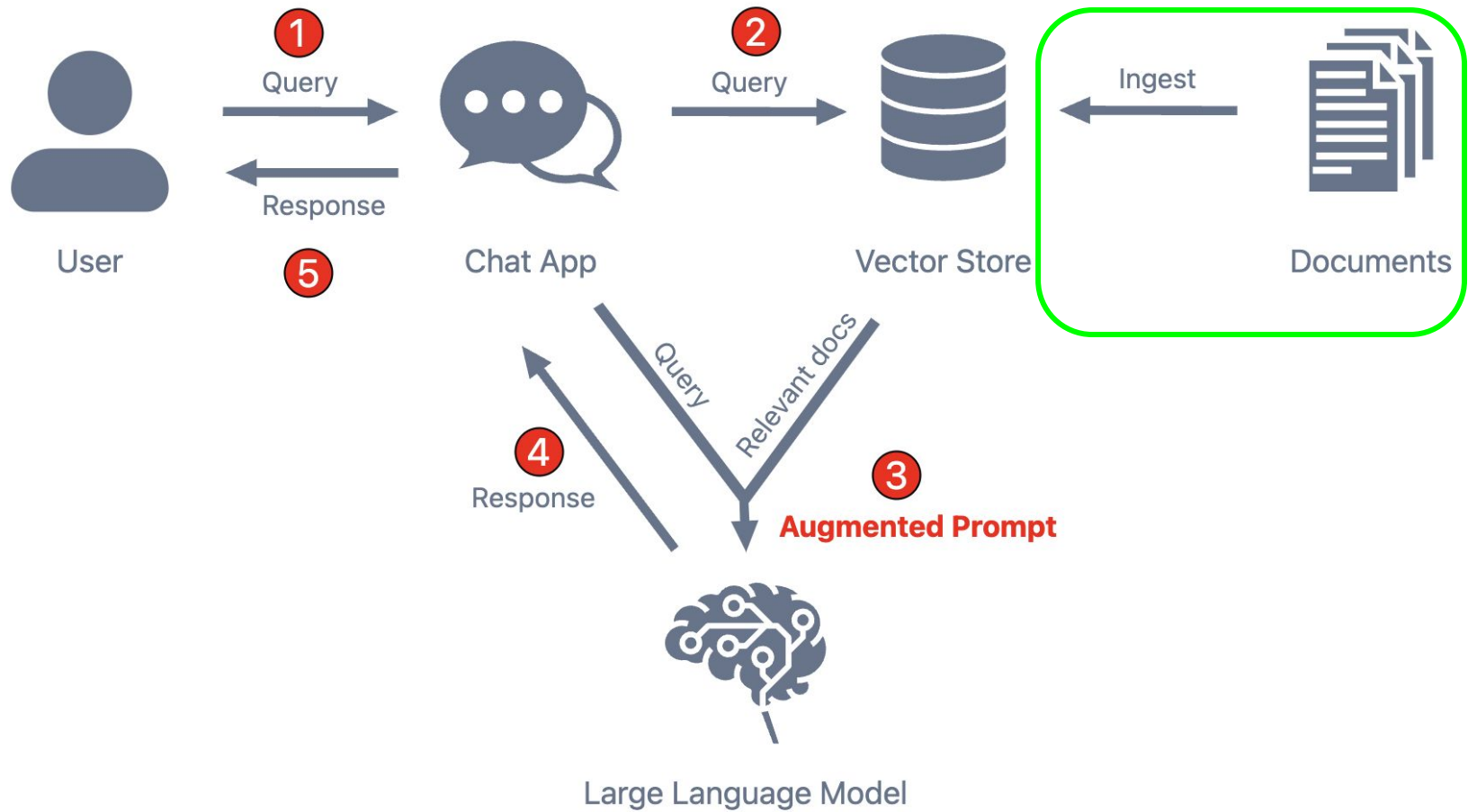
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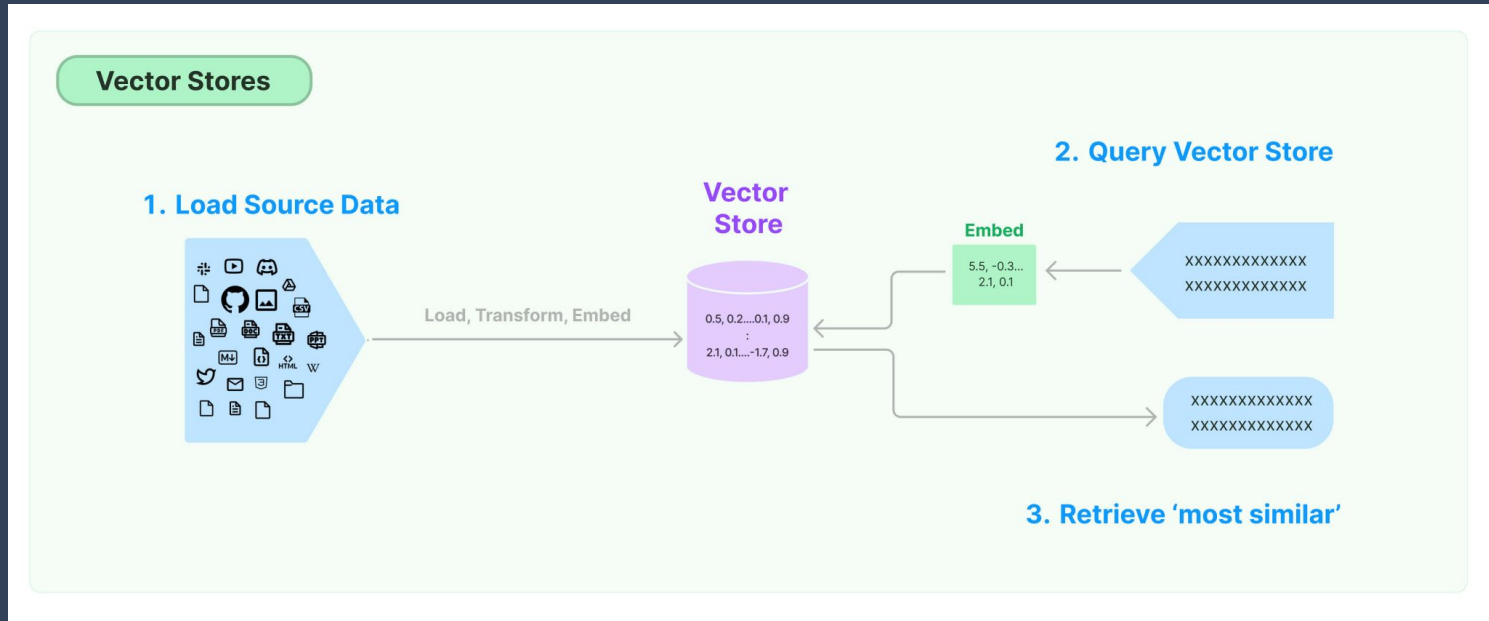
We will zoom in to different components!



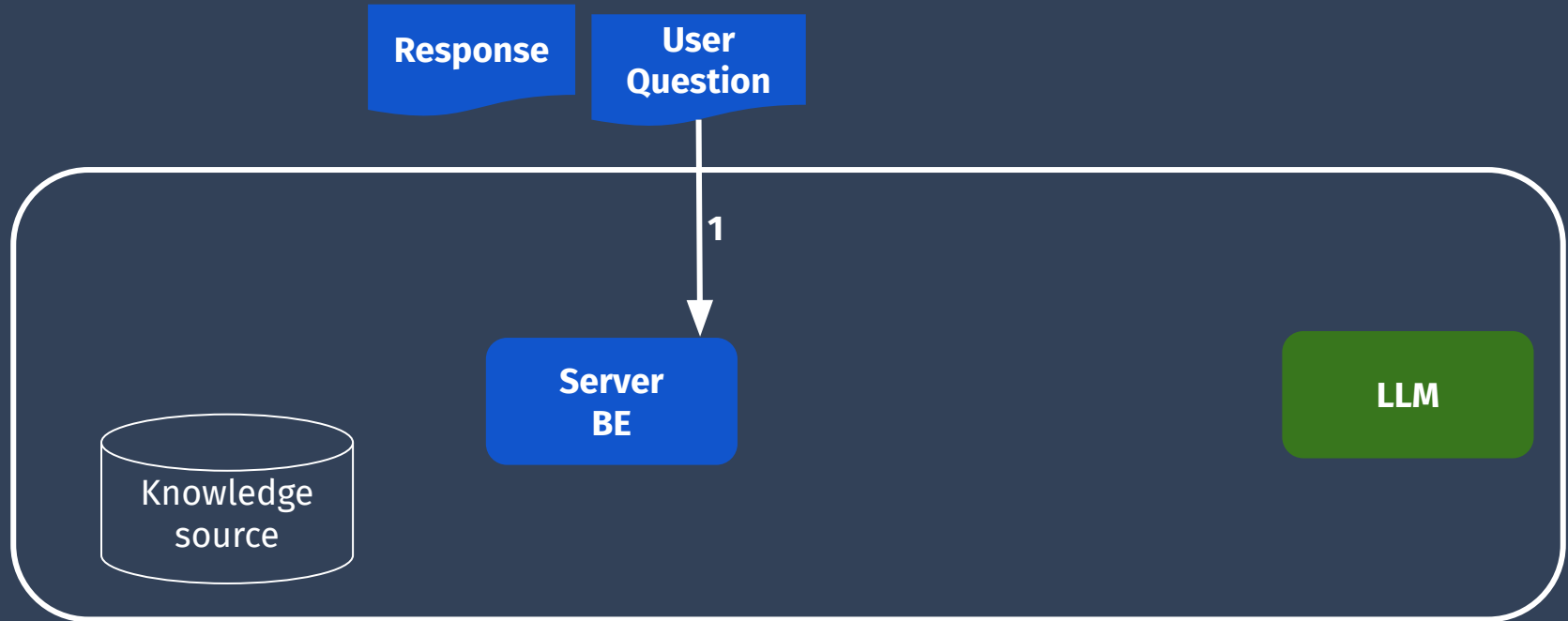




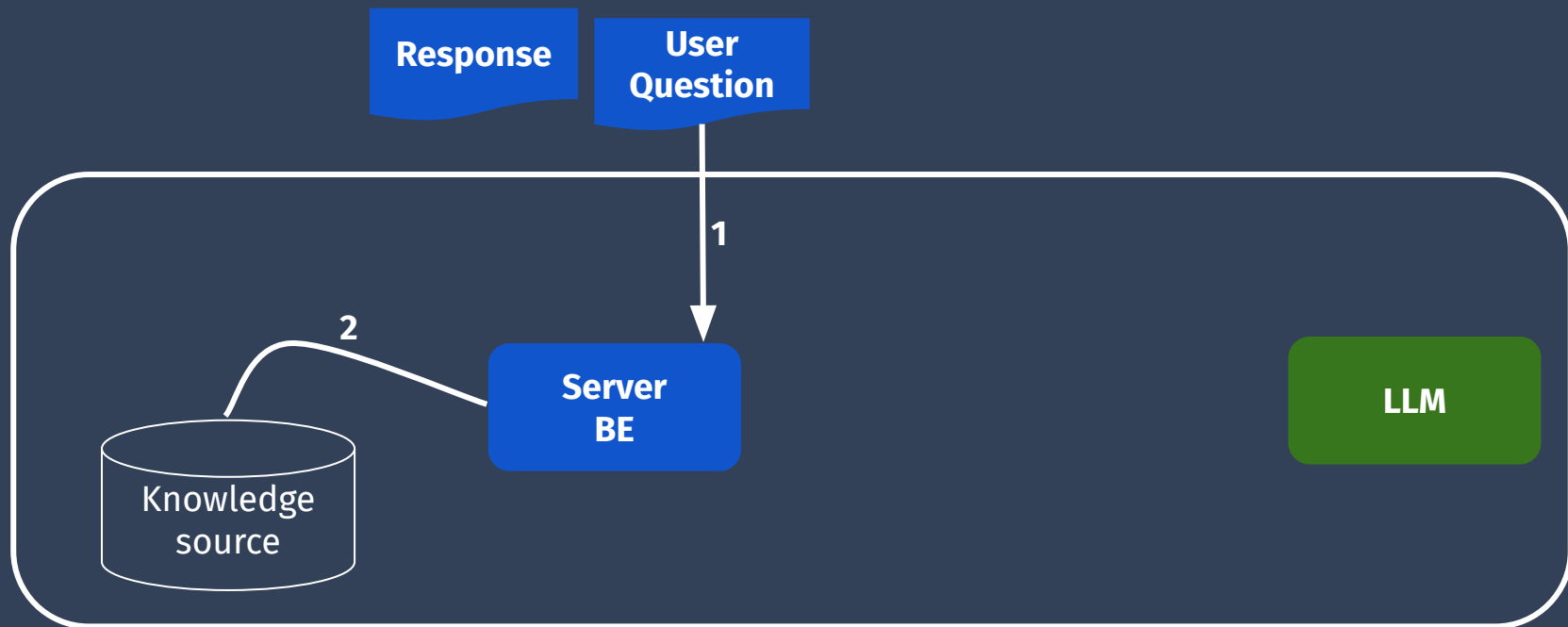
Vector Store



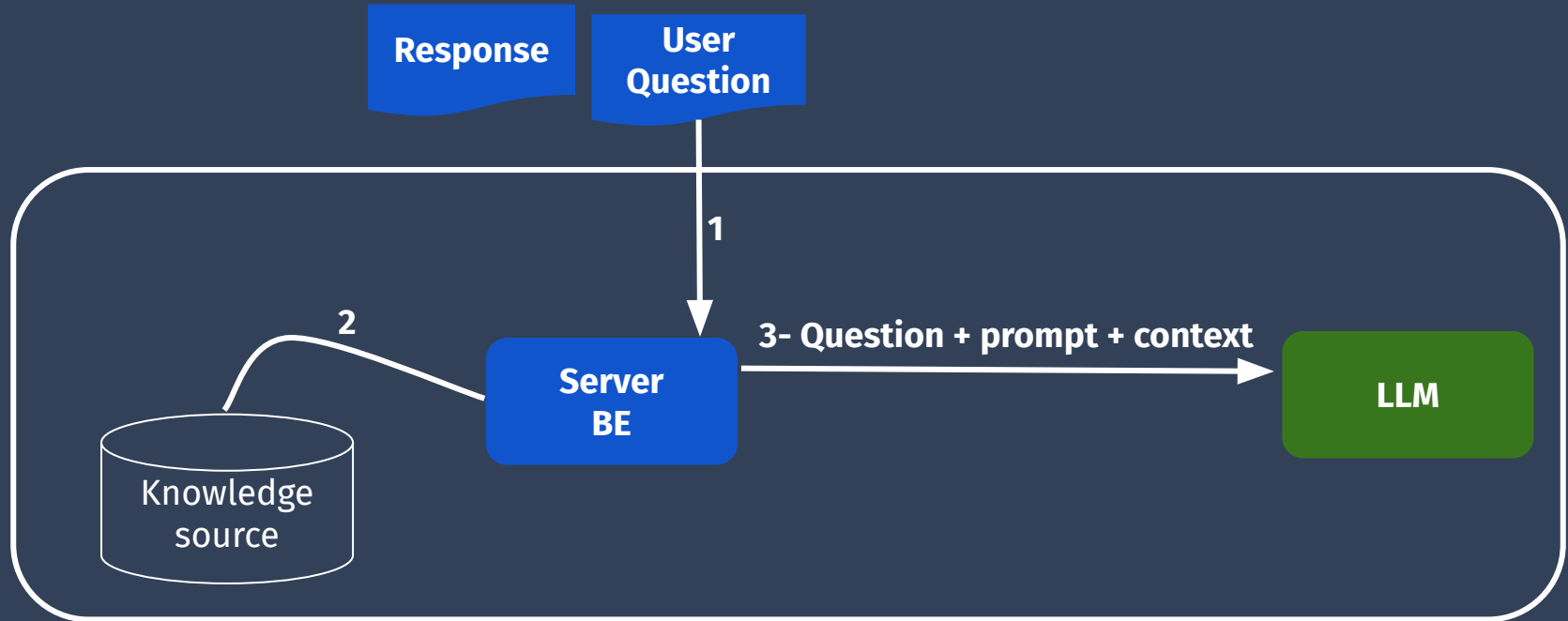
What a LLM service looks like?



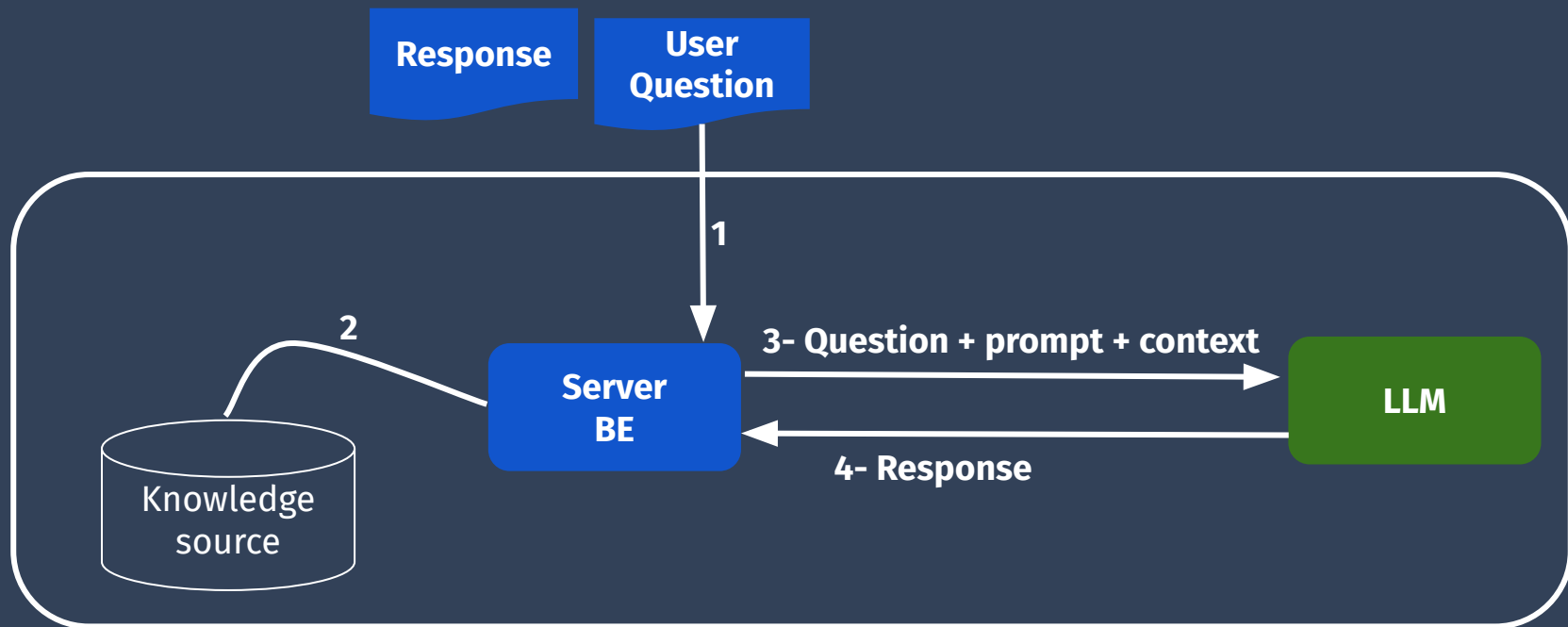
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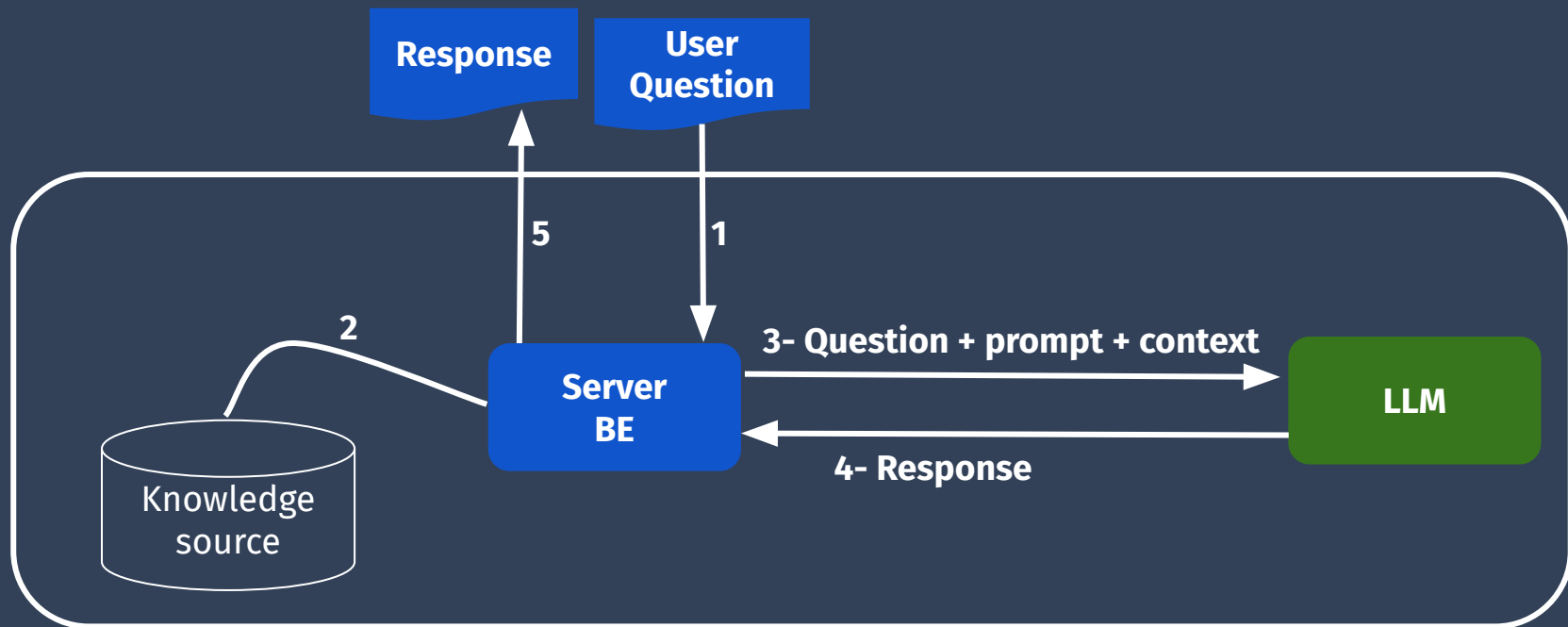
What a LLM service looks like?



What a LLM service looks like?



What a LLM service looks like?



Implementation

Agents

Agents:

- What is an agent?
- Why do we need them?



Agents: Why do we need them?

For a question like

“What's the best time to visit the Grand Canyon?”

Agents: Why do we need them?

For a question like

“What's the best time to visit the Grand Canyon?”

Although LLM has common knowledge. To answer this question it may benefit from:

- Access to latest weather predictions and patterns
- Tourism seasons (and particular tourism data from this year)
- Other factors that influence the visitor experience.
- ...

What do we need to add?

Agents: Why do we need them?

For a question like

“What's the best time to visit the Grand Canyon?”

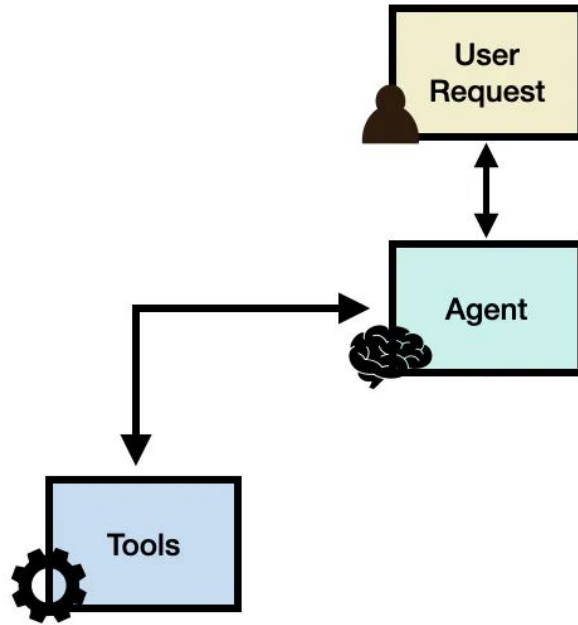
Although LLM has common knowledge. To answer this question it needs:

- Access to whether predictions and patterns
- Tourism seasons
- Other factors that influence the visitor experience.
- ...

What do we need to add?

Tools

Agents with Tools



Agents with Tools

- Tools at the disposal of the LLM that can assist it in finishing a task
- How the LLM would choose these tools?
 - Which prompting technique help with that?
 - More on this later

Agents: Why do we need them?

For a question like

User: What are good places to visit in Arizona?

System: Grand Canyon is a very common attraction that ...

User: What's the best time to visit it?

System: The best time to visit Grand Canyon is ...

...

User: What is the average cost of doing that

...

What do we need to add?

Agents: Why do we need them?

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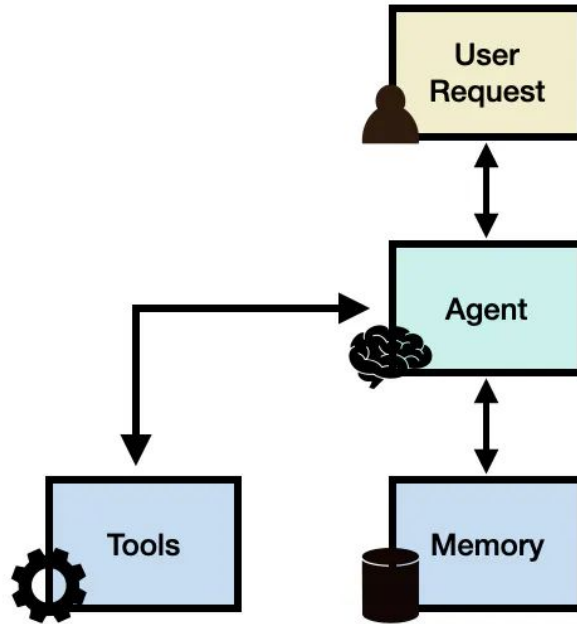
What do we need to add?

Memory

Agents: Memory

- Short-term memory
 - Context information about the agent's current state
 - Determined by in-context learning which means it is short and finite due to context window constraints.
- Long-term memory
 - Agent's past thoughts, actions, and observations that need to be retained and recalled over an extended period of time
 - External vector store: scalable retrieval

Agents with Tools & Memory



Agents: Why do we need them?

For a question like

“I need to search Apple Remote and find the program it was originally designed to interact with”

What do we need to add?

```
Thought 1: I need to search Apple Remote and find the
program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in
October 2005 by Apple ... originally designed to control the
Front Row media center program ...

Thought 2: Apple Remote was originally designed to control
the Front Row media center program. I need to search Front
Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row
Seat to Earth', 'Front Row Motorsports', 'Front Row
(software)', ...]

Thought 3: Front Row is not found. I need to search Front
Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple
Remote or the keyboard function keys. So the answer is
keyboard function keys.
Act 4: Finish[keyboard function keys]
```



Agents: Why do we need them?

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Planning

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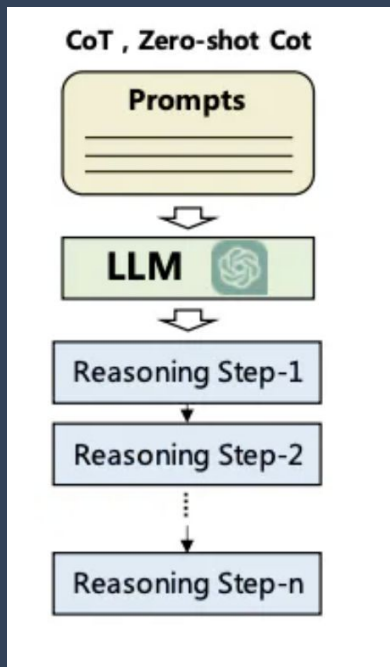
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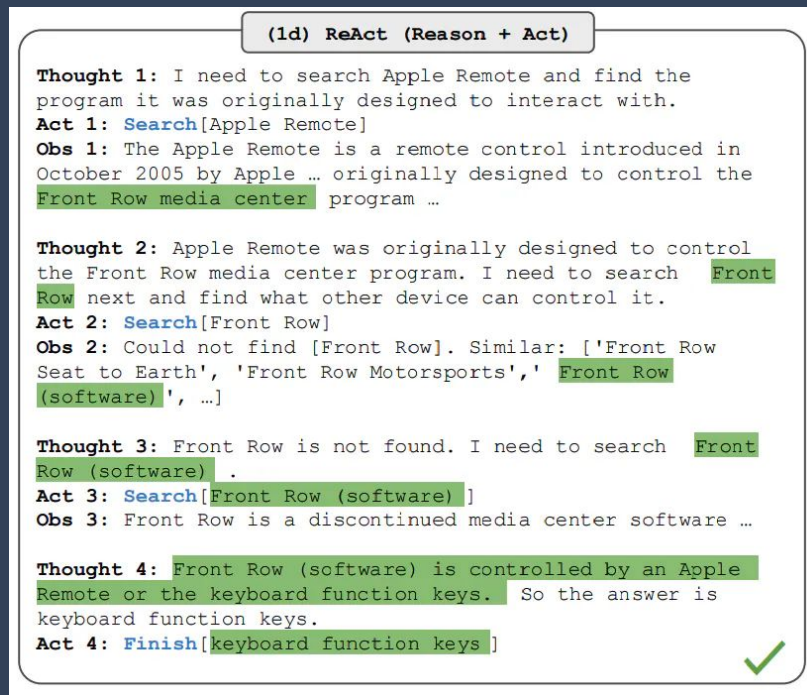


Agents: Planning

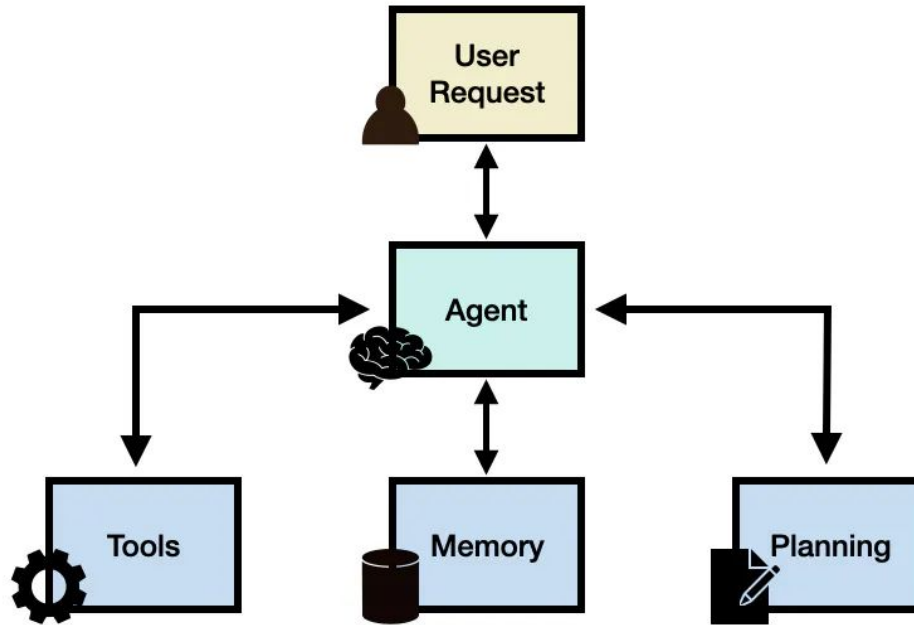
Planning without Feedback



Planning with Feedback



Agents: Putting it all together



Implementation

